Zariski Geometry: When Functions Become the Space

N-20241026

§1.1 Which geometry can a polynomial equation remember?

Consider the two equations $x = 0$ and $x^2 = 0$ over a field k. They have the same solutions in k, but the quotient algebras

$$k[x]/(x) \cong k, \qquad k[x]/(x^2)$$

are different: the class ϵ of x in the second ring is nonzero with $\epsilon^2 = 0$. Before defining any topology, we have encountered a choice. Do we study *ordinary point sets*, where these equations agree, or an algebraic space retaining infinitesimal equations, where they differ? This note follows the standard construction of affine geometry from rings, as organized in the *Stacks Project*, especially the chapters on spectra and affine schemes.[1]

First let k be algebraically closed, and put $A = k[x_1, \dots, x_n]$. For an ideal $I \subseteq A$, define

$$V(I) = \{a \in k^n : f(a) = 0 \text{ for all } f \in I\}.$$

Conversely, a subset $X \subseteq k^n$ determines

$$I(X) = \{f \in A : f(a) = 0 \text{ for every } a \in X\}.$$

The operations reverse containment: if $I \subseteq J$ then $V(J) \subseteq V(I)$. They also exchange natural operations:

$$V(I + J) = V(I) \cap V(J), \qquad V(IJ) = V(I) \cup V(J).$$

For the two axes, $V(x) \cup V(y) = V(xy)$, whereas $V(x) \cap V(y) = V(x, y) = \{(0, 0)\}$.

§1.2 What Points Forget in Coordinate Rings

The coordinate ring of an algebraic set X is

$$k[X] = A/I(X).$$

Its elements are polynomial functions restricted to X; two polynomials define the same function on X precisely when their difference belongs to $I(X)$.

Hilbert's Nullstellensatz is the crucial bridge. For an ideal I over an algebraically closed field,

$$\boxed{I(V(I)) = \sqrt{I} = \{f \in A : \text{some } f^m \in I\}.}$$

Thus point sets detect radical ideals. For instance, $I = (x^2)$ and $\sqrt{I} = (x)$ yield identical classical zero sets, and the associated coordinate ring of the *set* is $k[x]/(x)$. The thicker quotient $k[x]/(x^2)$ retains information that classical points have forgotten.

[1] Stacks Project, Tag 00DY, and Tag 01HR on affine schemes.

Example 1.2.1 (The curved hyperbola is an affine punctured line)
The parabola $Y = V(y - x^2)$ has

$$k[Y] = k[x, y]/(y - x^2) \cong k[x]$$

by eliminating y. The hyperbola $H = V(xy - 1)$ has

$$k[H] = k[x, y]/(xy - 1) \cong k[x, x^{-1}].$$

The projection $(x, y) \mapsto x$ identifies H with $\mathbb{A}^1 \setminus \{0\}$; its inverse is $t \mapsto (t, t^{-1})$. The identity can be checked in either direction and illustrates why the algebra of functions can reveal structure obscured by a drawing.

§1.3 Closed sets and the deliberately coarse topology

The closed subsets of affine space are the sets $V(I)$. They form a topology because finite unions and arbitrary intersections of algebraic sets are again algebraic. Over an infinite field, a nonzero polynomial in one variable has only finitely many roots, so a proper Zariski-closed subset of $\mathbb{A}^1$ is finite. Consequently nonempty Zariski-open subsets of the affine line have finite complements. This topology is useful for algebraic dependence; it does not try to measure Euclidean distance.

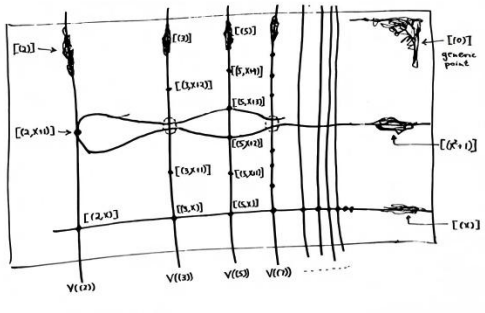


Figure 1.1: Geometric sketches can suggest shapes, while polynomial equations and their function rings determine the algebraic structure.

For $f \in k[X]$, the distinguished open set

$$D(f) = \{p \in X : f(p) \neq 0\}$$

is the region where division by f makes sense. In particular,

$$D(x) \subseteq \mathbb{A}^1, \qquad \mathcal{O}(D(x)) = k[x, x^{-1}].$$

This can be derived directly: localizing $k[x]$ at the multiplicative set $\{1, x, x^2, \ldots\}$ introduces fractions $g(x)/x^m$. Algebraic geometry uses these localizations in place of functions defined using arbitrary analytic neighborhoods.

§1.4 Sheaves, Stalks, and Gluing

To see how local data are organized, it is convenient to pass to the full spectrum. For any commutative ring R with 1,

$$\operatorname{Spec} R = \{\mathfrak{p} : \mathfrak{p} \text{ a prime ideal of } R\},$$

with closed sets $V(I) = \{\mathfrak{p} : I \subseteq \mathfrak{p}\}$ and basic opens $D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}$. The *structure sheaf* on this space has the decisive local formulas

$$\boxed{\mathcal{O}_{\operatorname{Spec} R}(D(f)) = R_f, \qquad \mathcal{O}_{\operatorname{Spec} R,\mathfrak{p}} = R_{\mathfrak{p}}.}$$

The second is the stalk at $\mathfrak{p}$: denominators are allowed exactly when they lie outside $\mathfrak{p}$.

Why is a sheaf necessary? If $D(f) = \bigcup_i D(f_i)$, a function described by sections $s_i \in R_{f_i}$ must satisfy

$$s_i|_{D(f_i f_j)} = s_j|_{D(f_i f_j)}$$

on every overlap. The sheaf condition asserts that compatible descriptions glue uniquely. For a finite standard-open cover of $\operatorname{Spec} R$, the algebraic content is the exact sequence

$$0 \longrightarrow R \longrightarrow \bigoplus_i R_{f_i} \longrightarrow \bigoplus_{i,j} R_{f_i f_j},$$

where the last arrow sends (s_i) to the differences of its restrictions. This is not merely a slogan about gluing: it says exactly what agreement on overlaps means and why a global section is determined by its local pieces.

As a small check, $D(t) \cup D(1-t) = \operatorname{Spec} k[t]$, since the ideal $(t, 1-t)$ contains 1. The two representations $t/t \in k[t, t^{-1}]$ and $(1-t)/(1-t) \in k[t, (1-t)^{-1}]$ agree on the overlap and glue to the global function 1.

§1.5 Branches and Infinitesimal Directions

Consider again

$$X = V(xy) \subseteq \mathbb{A}^2, \qquad R = k[x, y]/(xy).$$

Its two irreducible components correspond to the minimal primes (x) and (y). The open $D(x)$ has ring

$$R_x \cong k[x, x^{-1}],$$

because $xy = 0$ and x has become invertible, forcing $y = 0$. Similarly $R_y \cong k[y, y^{-1}]$. On these opens, only one branch remains. Their intersection $D(xy)$ is empty.

The origin, however, lies in neither $D(x)$ nor $D(y)$. Its local ring is

$$R_{(x,y)} = \big(k[x, y]/(xy)\big)_{(x,y)}.$$

The maximal ideal $\mathfrak{m} = (x, y)R_{(x,y)}$ satisfies

$$\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2,$$

since the linear classes of x and y survive independently; the relation $xy = 0$ only appears in degree two. A smooth curve has one-dimensional tangent space at a smooth closed point. Thus the local ring detects the crossing as singular, even though each punctured branch individually looks like a line.

The nilpotent example $k[\epsilon]/(\epsilon^2)$ makes a different point. Its spectrum has a single prime (ϵ), hence one underlying topological point, but its global sections have the form

$$a + b\epsilon, \qquad \epsilon^2 = 0.$$

Topology alone cannot distinguish it from $\operatorname{Spec} k$. The structure sheaf does. This is why the geometry of schemes retains both prime ideals as points and rings of local functions.

§1.6 Morphisms reverse arrows on rings

For affine schemes, a ring homomorphism $\varphi : B \to A$ induces a map of spectra

$$\operatorname{Spec} A \longrightarrow \operatorname{Spec} B, \qquad \mathfrak{p} \longmapsto \varphi^{-1}(\mathfrak{p}).$$

The inverse image of a prime ideal is prime, and the preimage of $D(b)$ is $D(\varphi(b))$; hence this map is continuous and compatible with local sections. Conversely, an affine-scheme morphism induces such a homomorphism on global functions. Schematically,

$$X \longrightarrow Y \quad \rightsquigarrow \quad \Gamma(Y, \mathcal{O}_Y) \longrightarrow \Gamma(X, \mathcal{O}_X).$$

One can see this without any categorical machinery. The map

$$\mathbb{A}^1 \longrightarrow \mathbb{A}^2, \qquad t \longmapsto (t, t^2)$$

pulls the coordinate functions x, y back to t, t^2; the induced ring map is

$$k[x, y] \longrightarrow k[t], \qquad x \mapsto t, \quad y \mapsto t^2,$$

whose kernel is $(y - x^2)$. This single calculation connects the geometry of the parabola, its coordinate ring, and the reversal of arrows. Localizations and stalks make the same correspondence meaningful near every point.